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DATA MODELING

Reading: Chapter 7

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Data Modeling

- A technique for modeling data
- We've covered
  - Entity Relationship Diagram (ERD)
  - DB Model (e.g., Access Design View)
- The goal is to identify the structure of data to be stored in the database
- Data model is not concerned with how it is stored in the DB
- Data modeling is a partnership between the client and analyst

In industry, the job title is usually Business Analyst

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The Building Blocks

- Entity
- Attribute
- Relationship
- Identifier

Entity name

Shares

code
name
price
quantity
dividend
pe

attributes

Remember our notation is slightly different from the text (e.g., plural entity name in upper camel case and singular attribute names in lower camel case)

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A Well-Formed Data Model

- Follow organization (e.g., your company) convention
- No ambiguity
  - All entities, attributes, relationships, and identifiers are defined
  - Names are meaningful to the client

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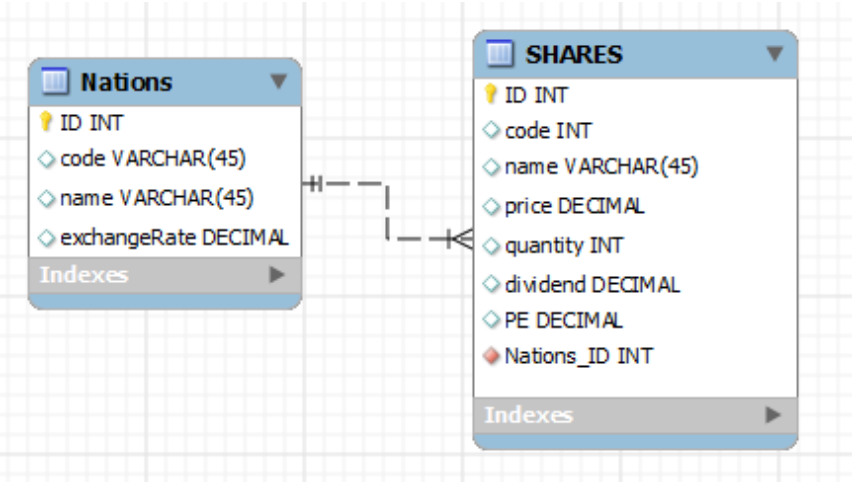
Relationships

- Shows relationships between entities
  - 1-1
  - 1-many
  - Many-many
  - Recursive
- ERD usually shows the relationship, not the implementation of the relationship (i.e., no foreign keys)



Workbench uses more of a DB model style (not an ERD)

ERD Iteration

- You will usually iterate on your ERD model, as you learn the entities and relationships better
  - Is the level of detail correct?
  - Do the names represent the real-world entities and attributes?
  - Are the model components situated so that the design is clear (e.g., no hidden or crossing lines)
  - Are all exceptions handled?
  - Is the model accurate?

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Are We on Track

- Think about your solution to Q#1 from the mid-term exam
- Write an ERD that provides a design for the following database statement:
  The DB should store details about U.S. Presidents and their terms in office. Term in office can be limited to start year and end year. The DB should record details of their date of birth, gender and political party affiliation. Presidents are often associated with recent presidents in the era, so the DB should record the predecessor and successor of any president. The database should handle non-consecutive terms and should handle presidents whose political party changes.

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Are We on Track

- Sketch a solution (in a group)
- Use a simple electronic tool (MS Word), and start with the ERD-Template.docx on the class Web site
- Be prepared to show your solution to the class (and iteratively refine the solution)

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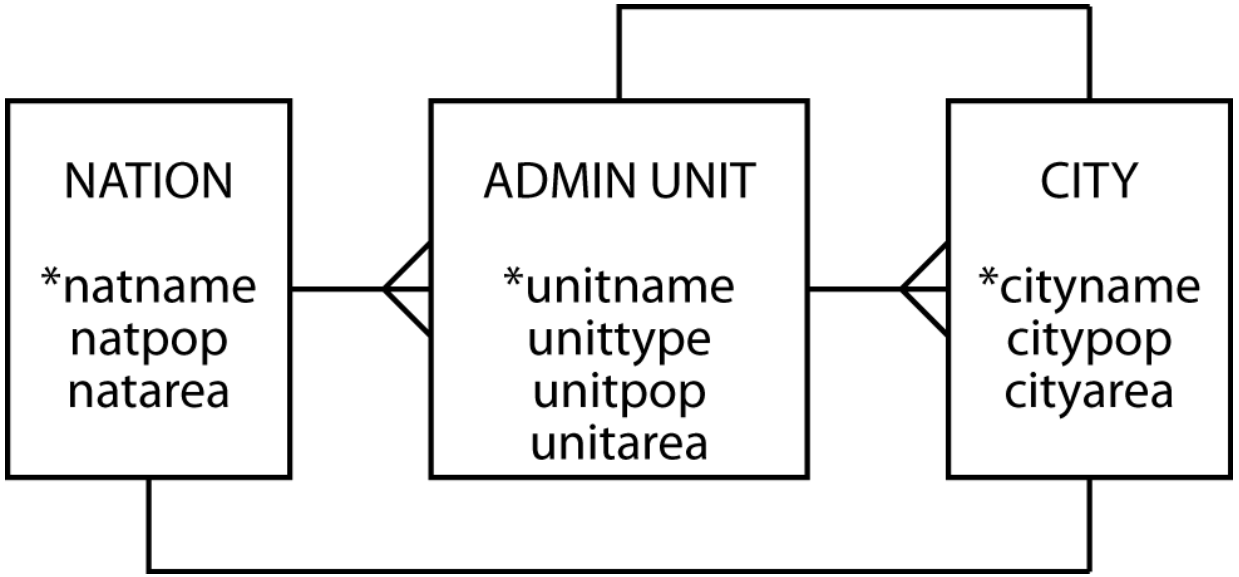
Pure geography



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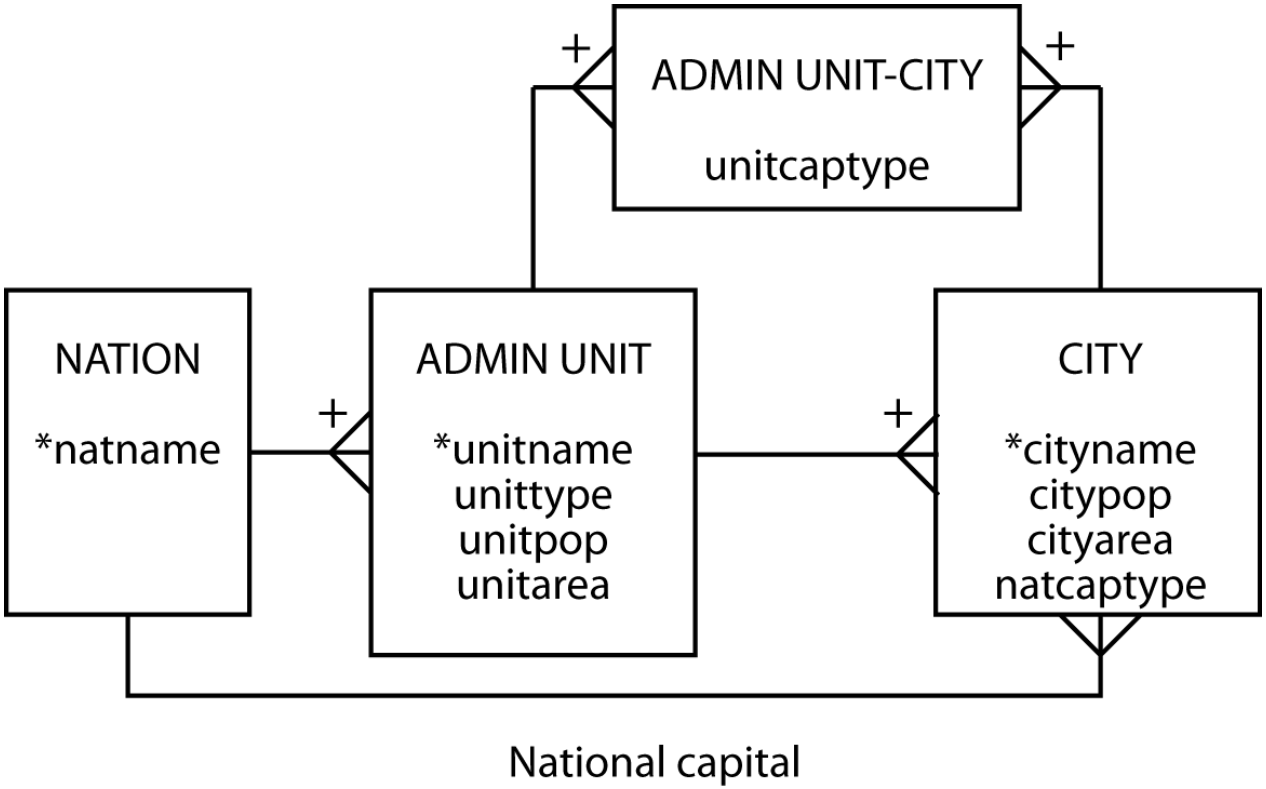
Geography revised



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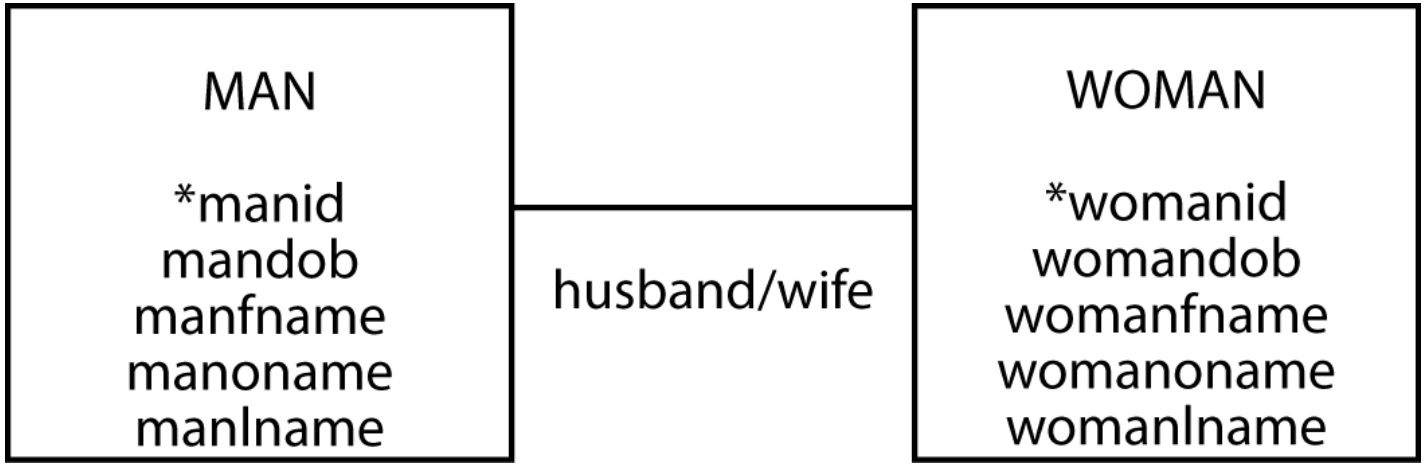
Family matters - take 1



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Family matters - take 2

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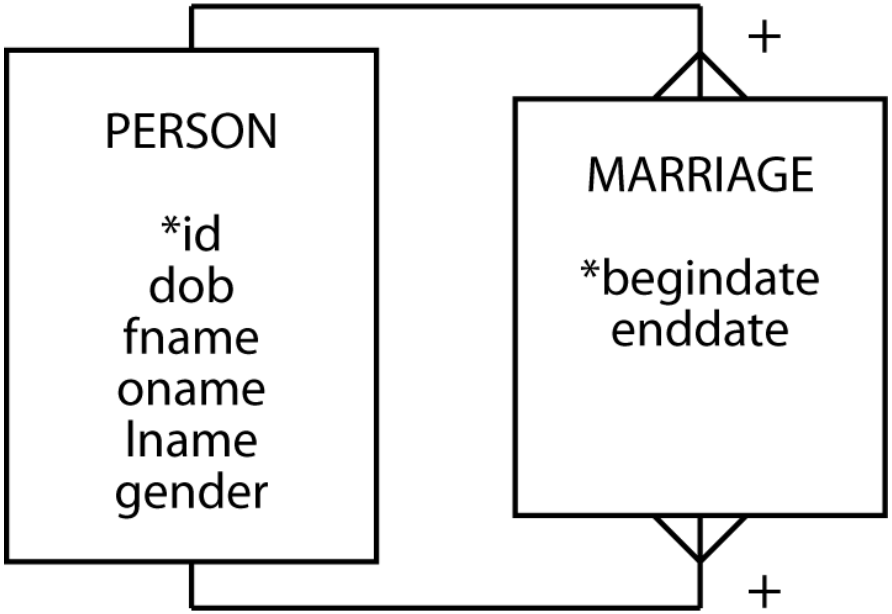
Family matters - take 3



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Family matters - take 4

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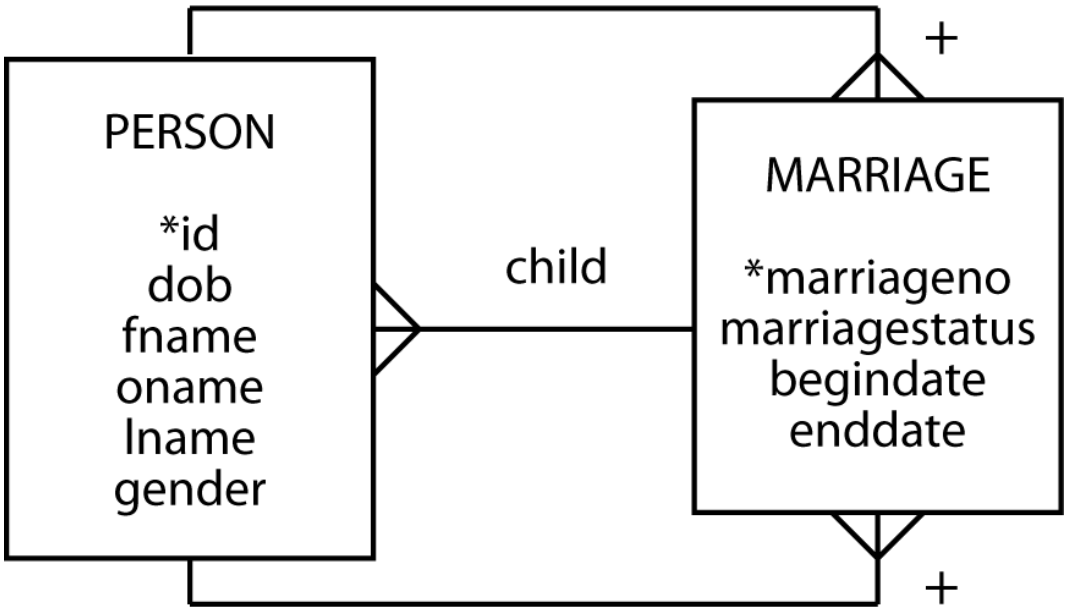
Family matters - take 5



Bookish matters - take 1

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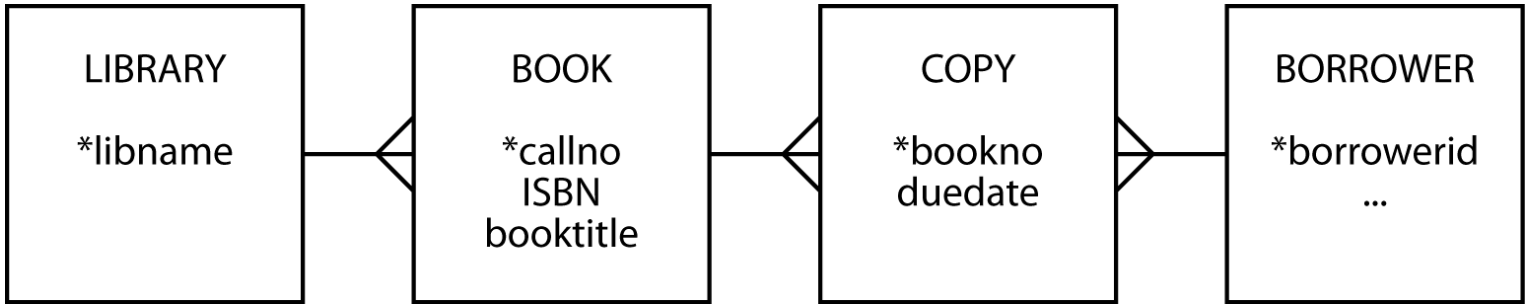
Bookish matters - take 2



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History - take 1

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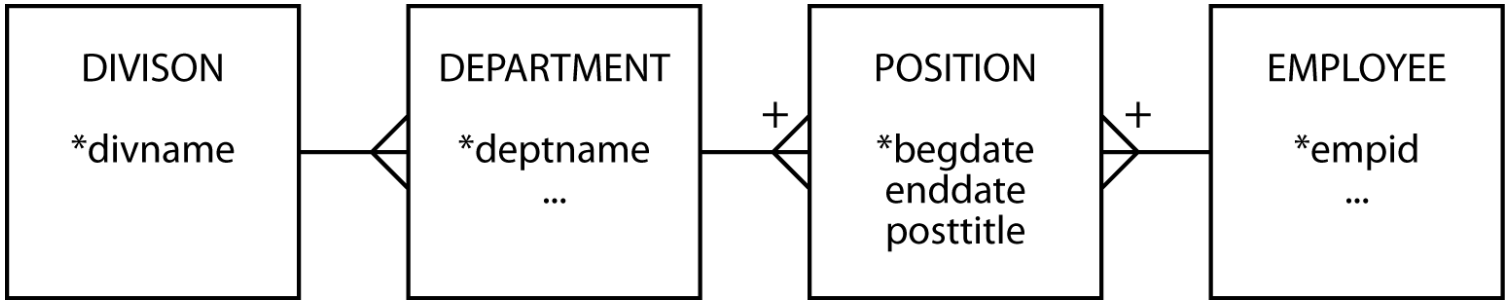
History - take 2



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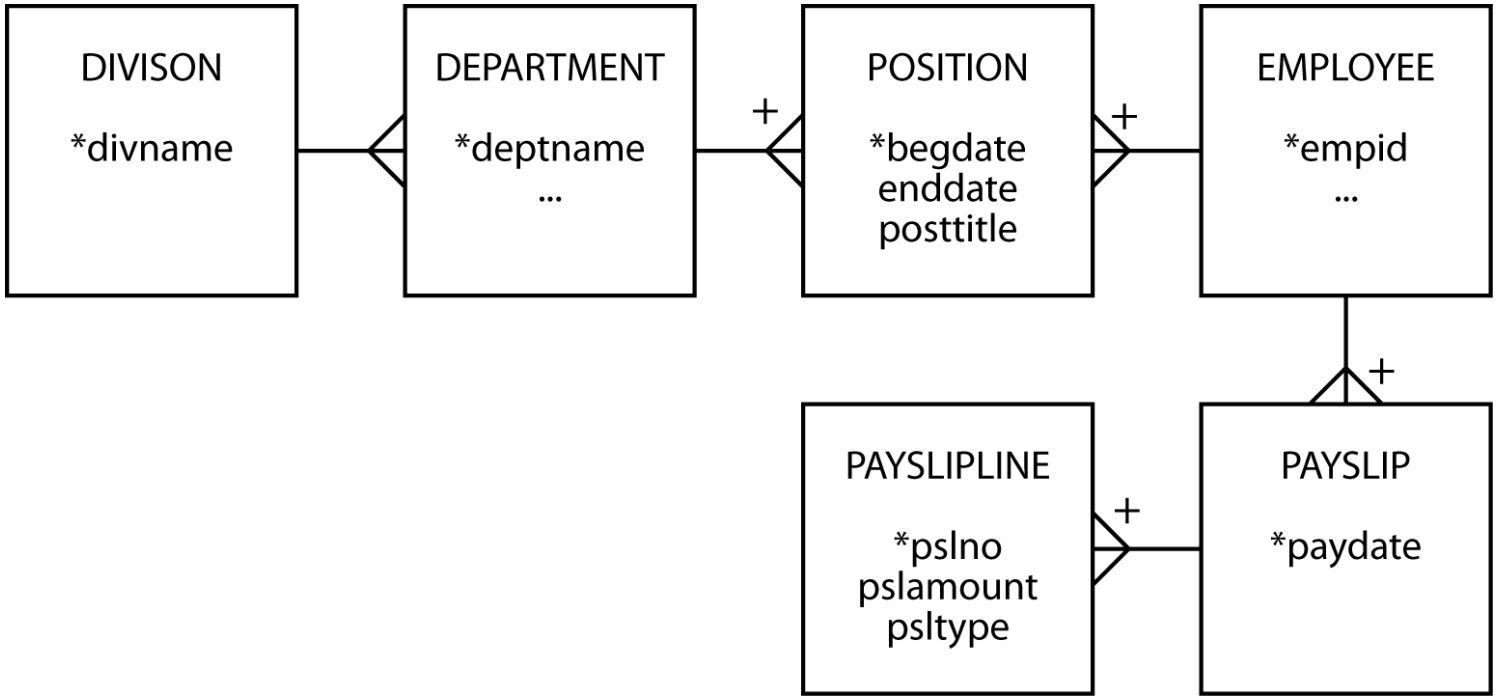
History - take 3



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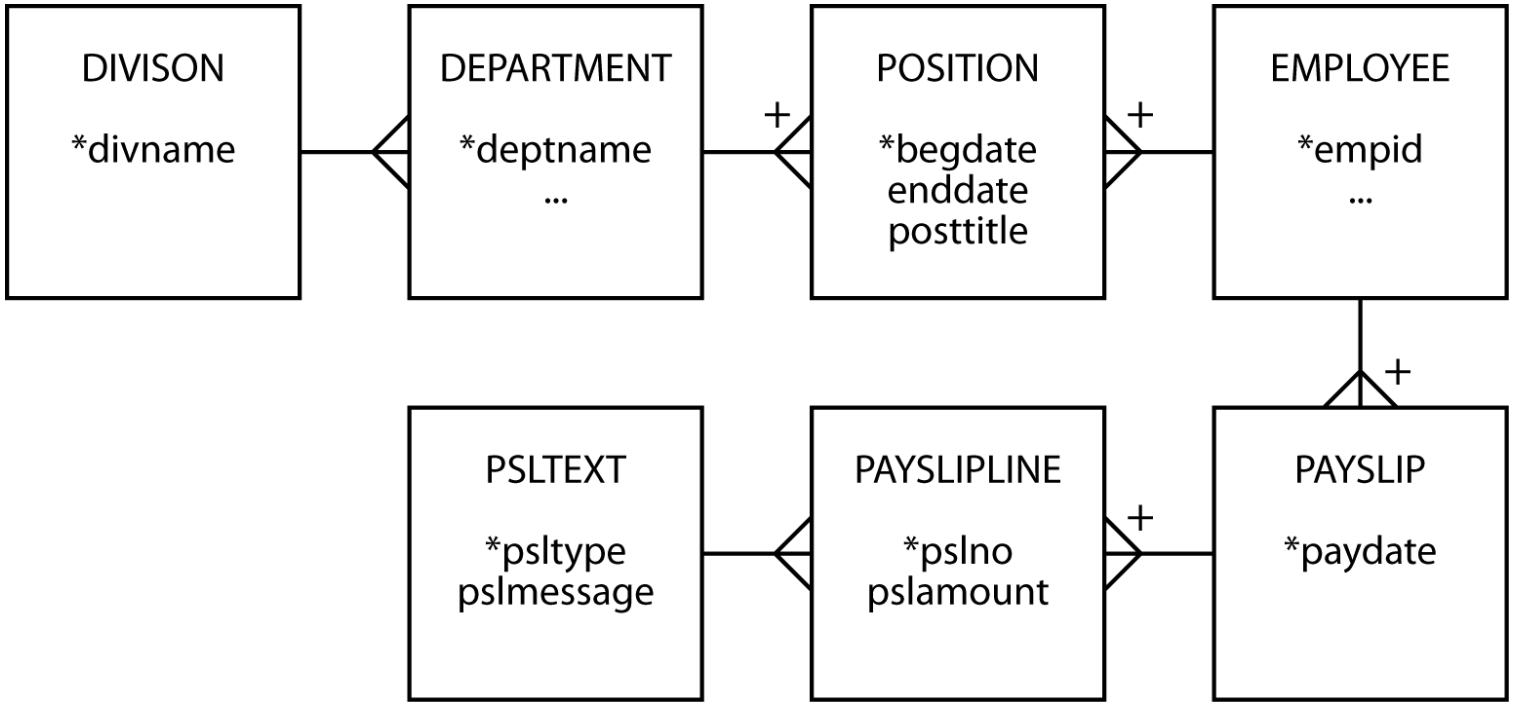
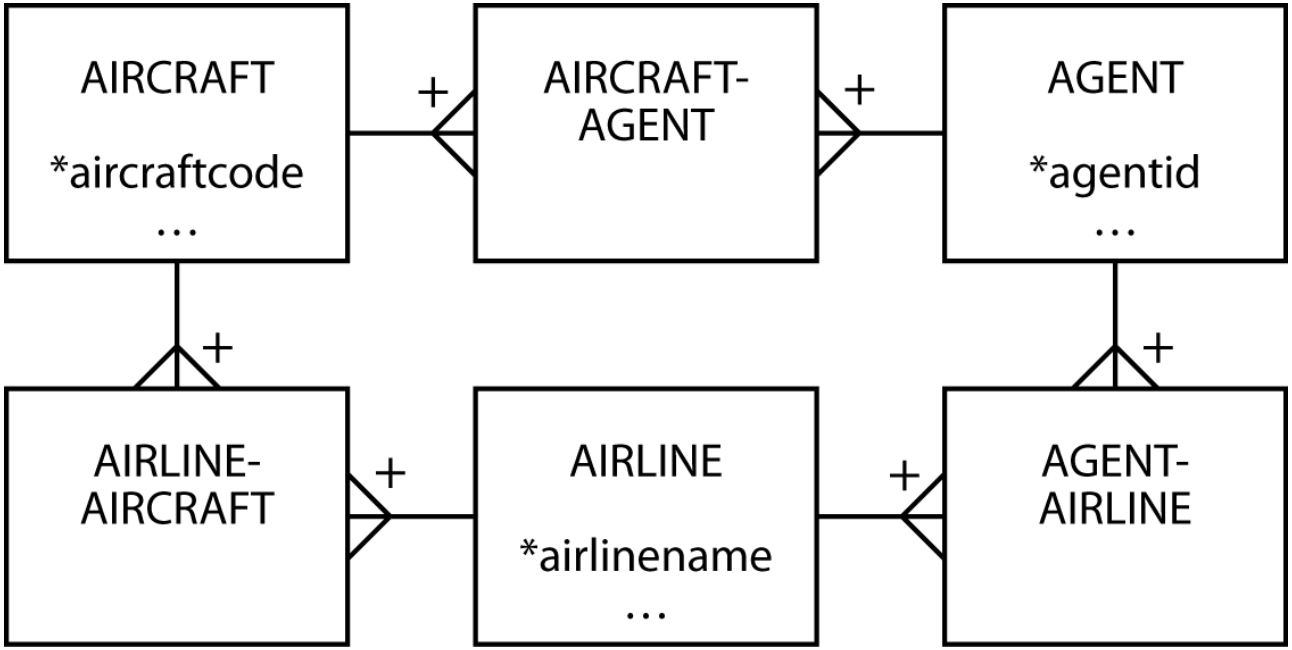
History - take 4



A ménage à trois for entities - take 1



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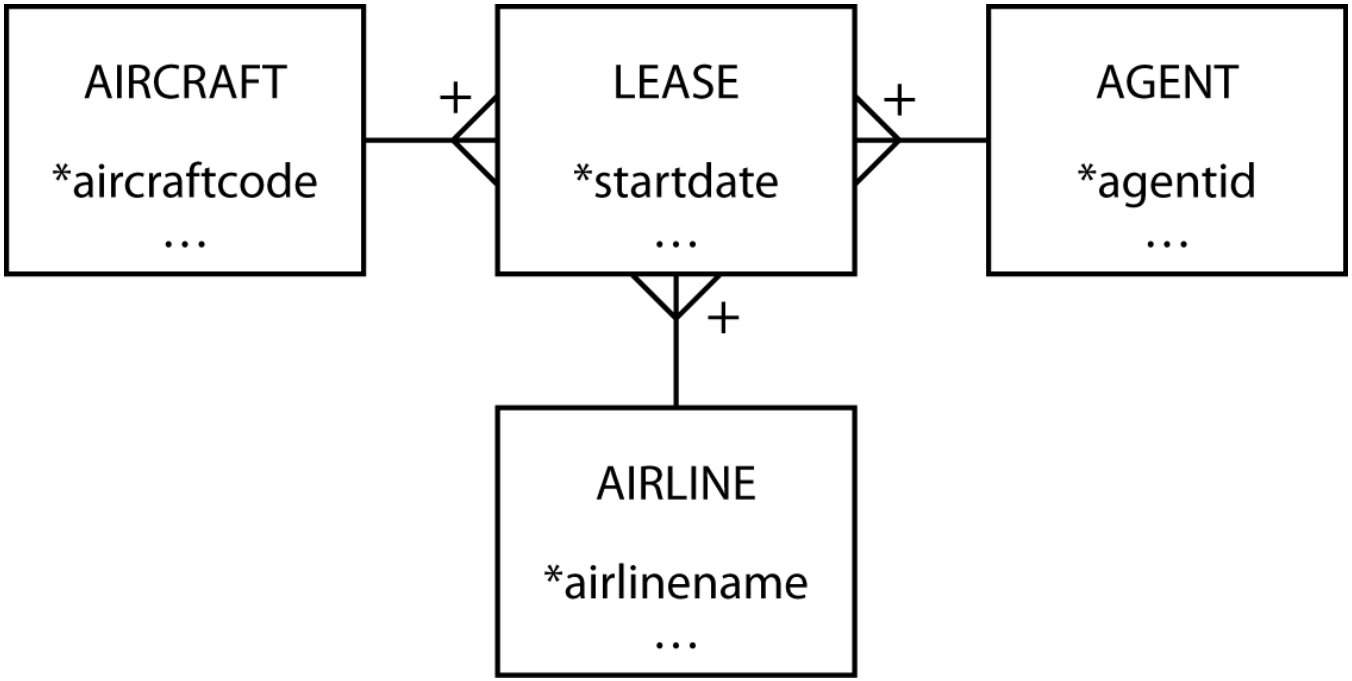
A ménage à trois for entities - take 2



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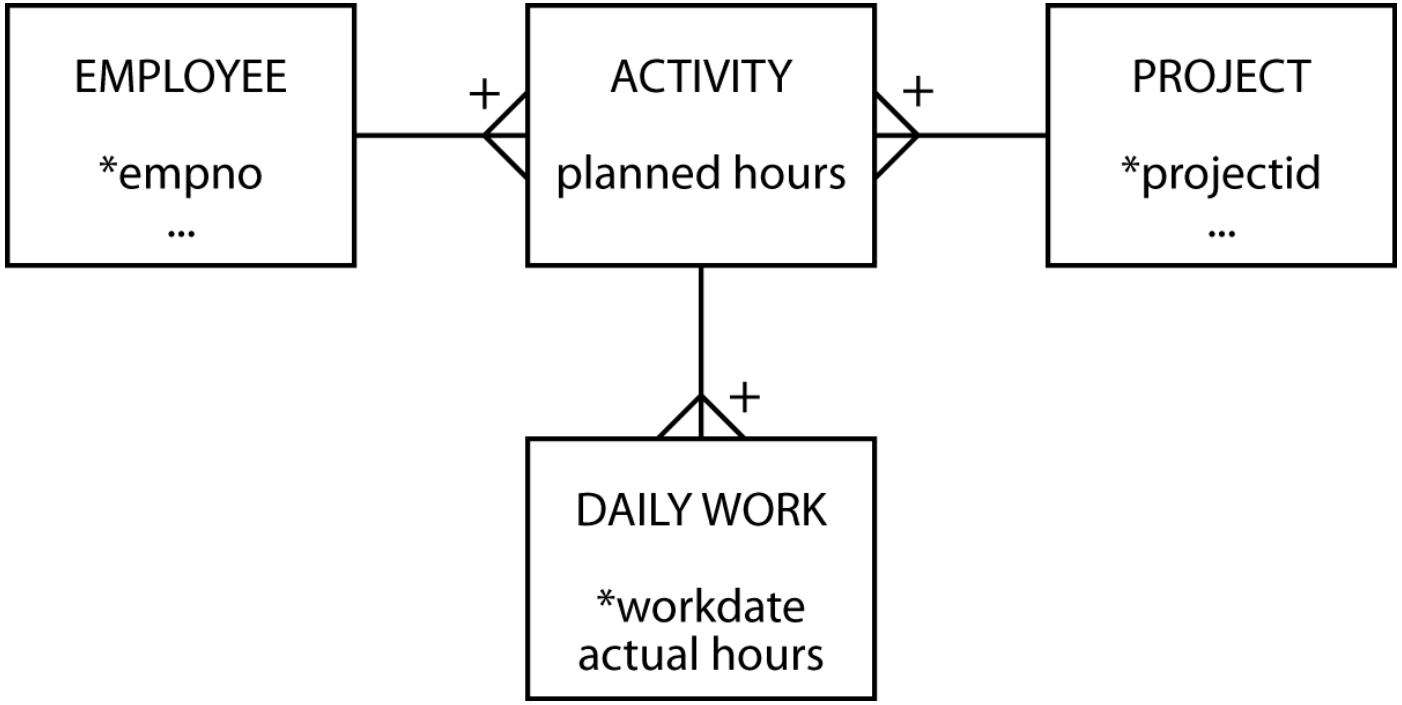
Planning and doing - take 1



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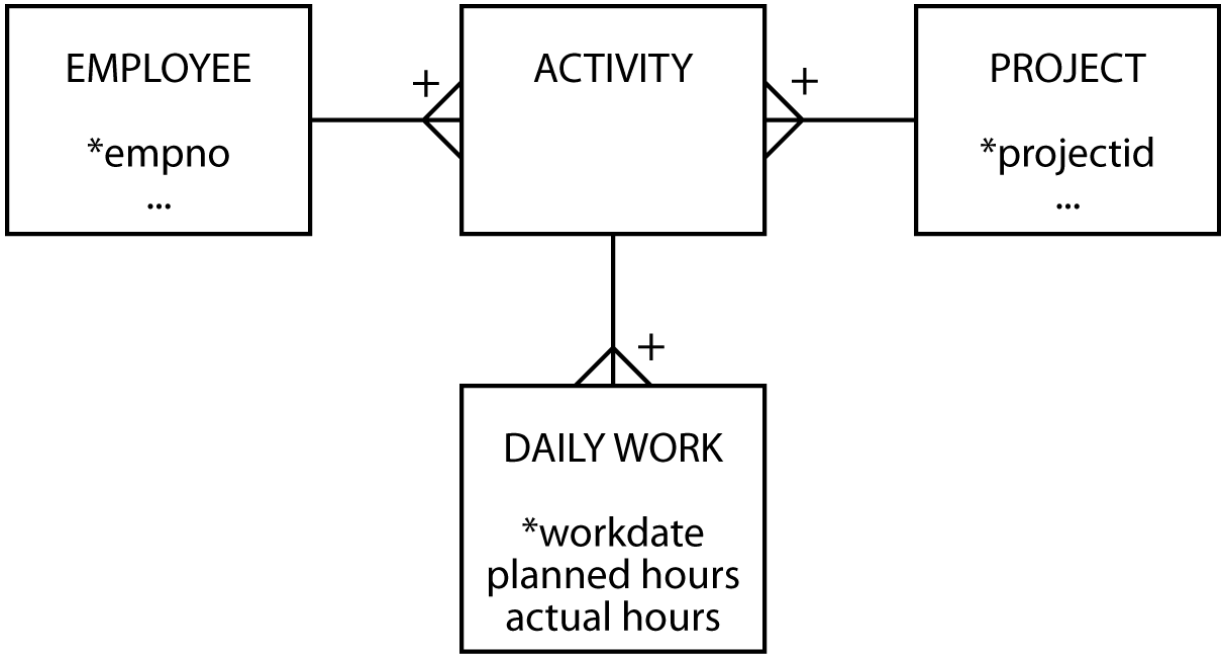
Planning and doing - take 2



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Entity types

- Independent
- Dependent
- Associative
- Aggregate
- Subordinate

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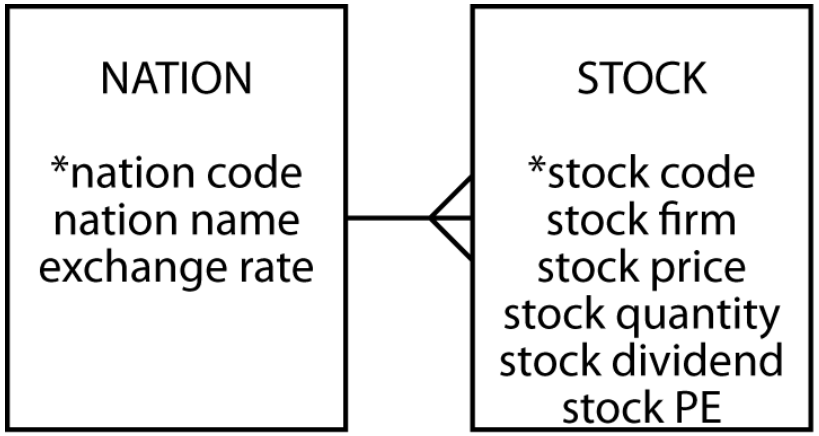
Independent

- Often a starting point
- Prominent in the client's mind
- Often related to other independent entities



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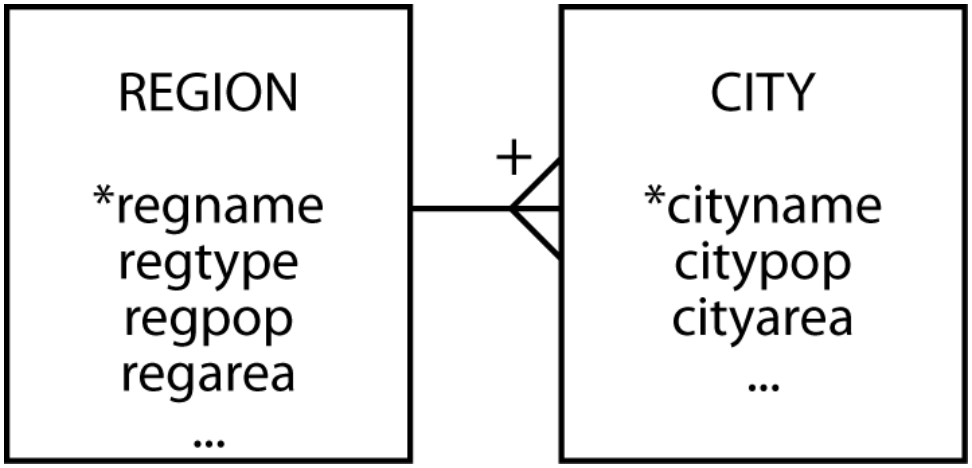
Dependent

- Relies on another entity for its existence and identification
- Can become independent if given an arbitrary identifier



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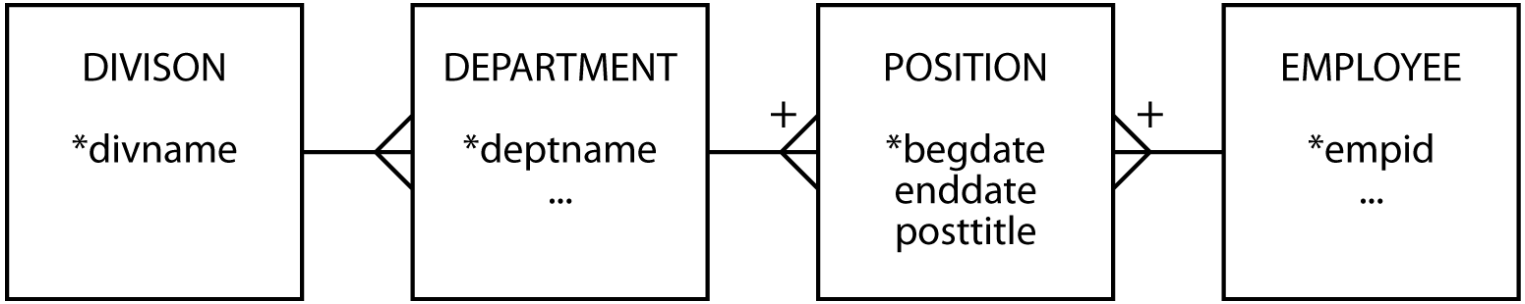
Associative

- A by-product of an m:m relationship
- Typically between independent entities
- Can store current or historical data
- Can become independent if given an arbitrary identifier



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Aggregate

- Created from several different entities that have a common prefix or suffix
- Commonly used with addresses or names

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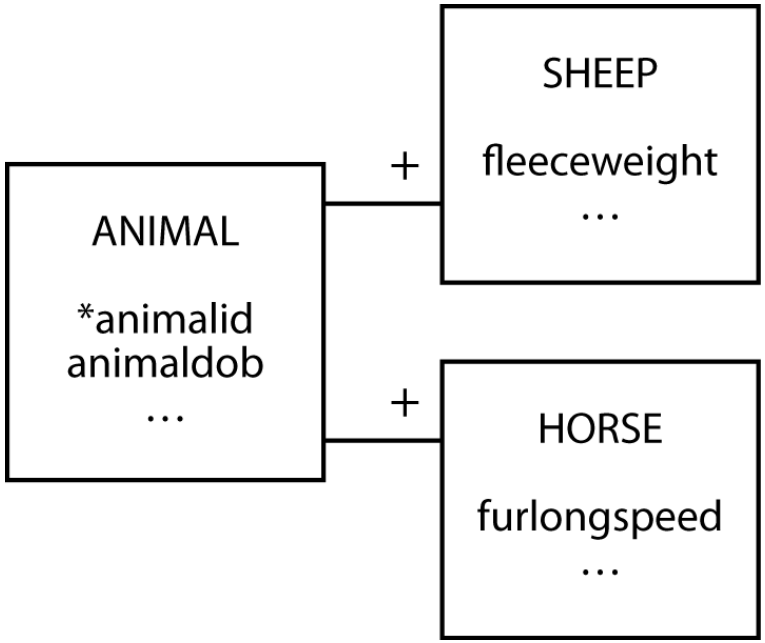
Subordinate

- An entity with data that can vary among instances



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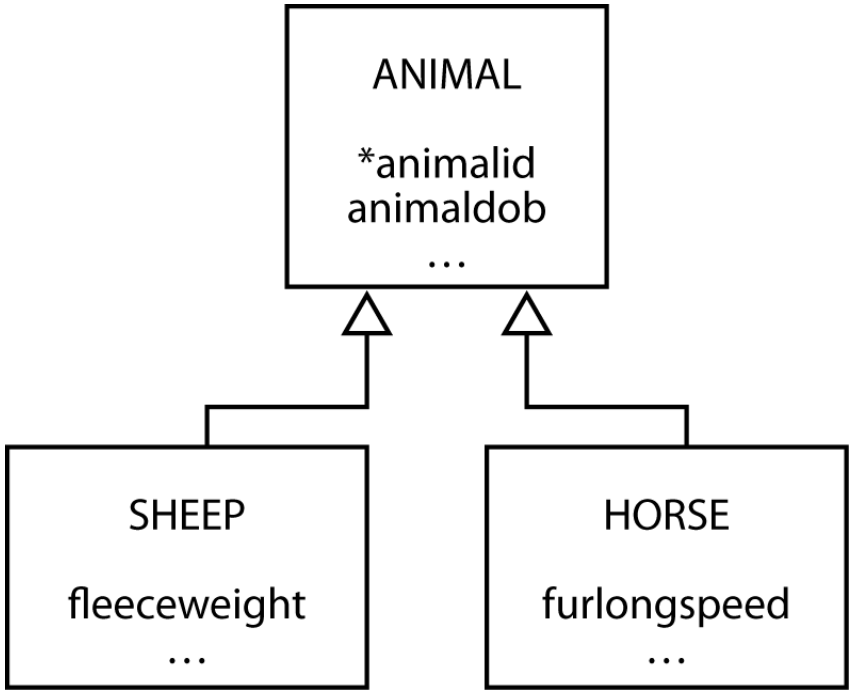
Generalization

- A relationship between a more general element and a more specific element



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Generalization

- Map with one table for each entity
- For each of the subtype entities the primary key is that of the supertype entity
- You must also make this column a foreign key so that a subtype cannot be inserted without the presence of the matching supertype

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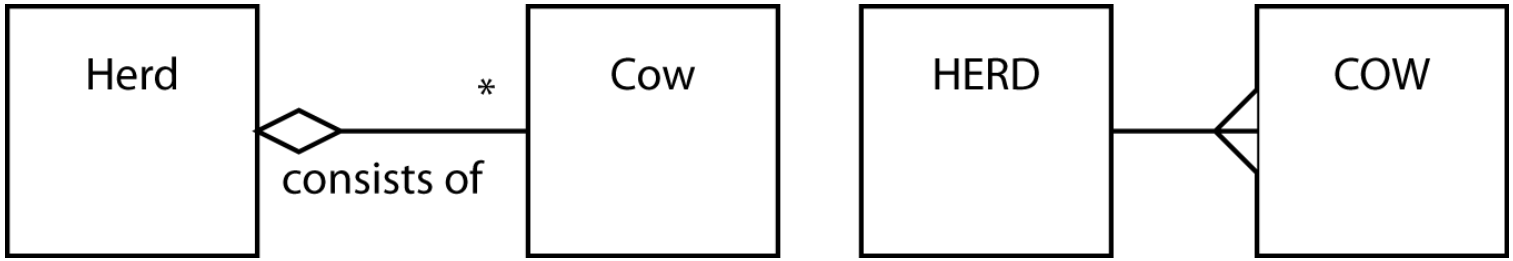
Aggregation

- Aggregation is a part-whole relationship between two entities



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Shared aggregation

- One entity owns another entity, but other entities can own that entity as well

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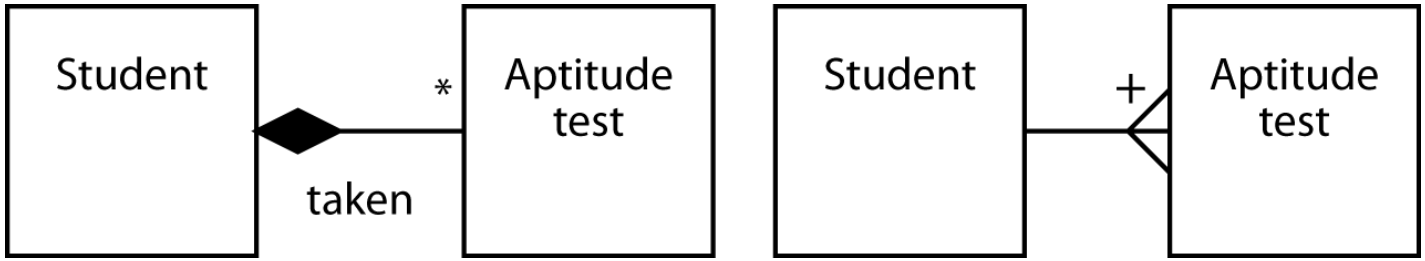
Composite aggregation

- One entity exclusively owns the other entity



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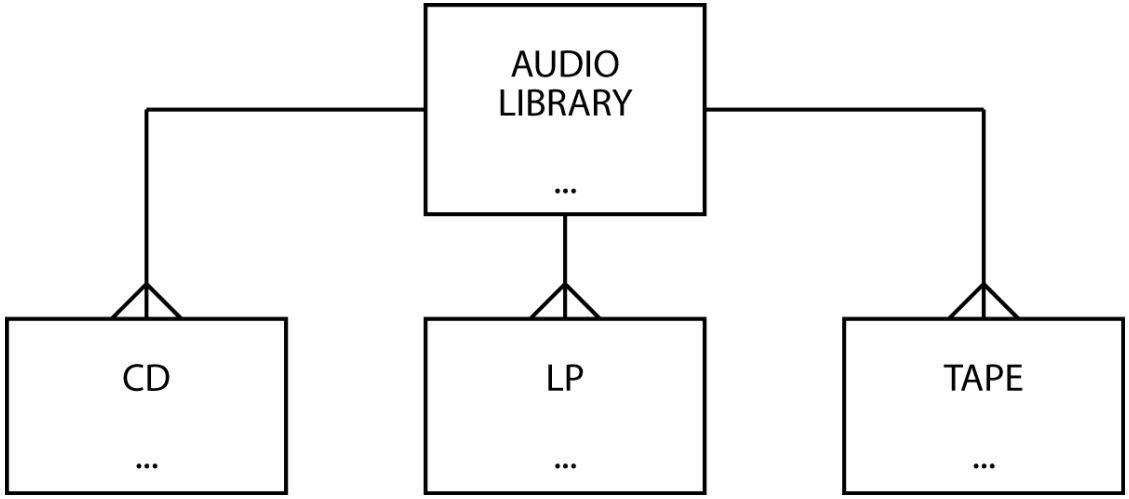
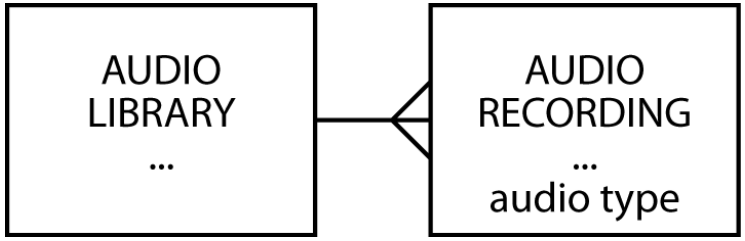
Data model contraction





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Hints on data modeling

- The model will expand and contract
- Invent identifiers where necessary
- Identifiers should have only one purpose – identification
- A data model does not imply ordering
- Create an attribute if ordering of instances is required
- An attribute's meaning must be consistent

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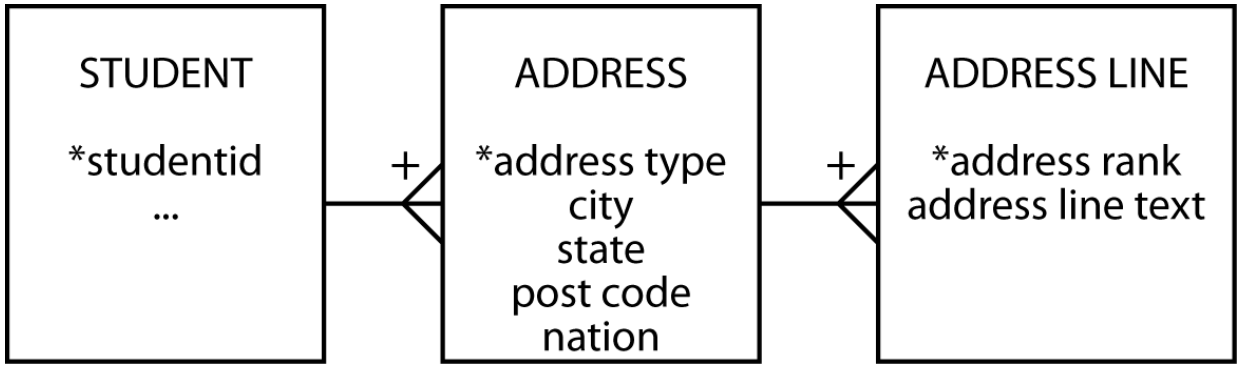
Names and addresses

- The query test
  - If an attribute has parts, are any of the parts ever likely to appear in a query?
- Have an understanding on representing names and addresses in a data model



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Hints on data modeling

- Single instance entities are OK
- Select names carefully
- Synonyms—different words have the same meaning
  - Get clients to settle on a common word or use views
- Homonyms—same word has different meanings
  - Clarify to avoid confusion
- Naming associative entities
  - Concatenate entity names if there is no obvious real world name

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Hints on data modeling

- Uncover all exceptions
- Label relationships to avoid ambiguity
- Keep the data model well-formed and accurate

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Meaningful identifiers

- An identifier is meaningful when some attributes of the entity can be inferred from the identifier's value

Advantages	Disadvantages
Recognizable and rememberable	Identifier exhaustion
Administrative simplicity	Reality changes
	Loss of meaningfulness

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Recommendation

- Nothing, however, is lost and much is gained by using **non-meaningful** identifiers
- Non-meaningful identifiers serve their sole purpose well
  - To uniquely identify an entity
- Attributes are used to describe the characteristics of the entity
- A clear distinction between the role of identifiers and attributes creates fewer data management problems

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The seven habits of highly effective data modelers

- Immerse
- Challenge
- Generalize
- Test
- Limit
- Integrate
- Complete

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Key points

- A high-fidelity data model handles all exceptions
- Identifiers need only identify an instance
- Data modeling skills take time to develop

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